Universiteit Antwerpen
Faculteit Toegepaste Ingenieurswetenschappen

5 – Infrastructuur & Beveiliging

Lab organization

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REFERENCES ..... FOUT! BLADWIJZER NIET GEDEFINIEERD.

1 Introduction

In this lab, we focus on creating IP networks and specifically on the security of these networks.

The motivation to build security into networks is not far-fetched. Just think of the many cyberattacks, malware attacks for which ransoms are demanded from large companies, viruses, and more.

> **"There are only two types of organizations: Those that have been hacked and those that don't know it yet!" ~ John Chambers, former CEO of Cisco Systems**

Over eight lab sessions, you will build a network from the ground up, starting with a single device and ending with a segmented company network that is protected by firewalls and VPNs, and that you will test yourselves. Everything is done on real hardware: you cable the devices, configure them, break them, and fix them again.

The lab follows one red thread: **you cannot secure what you do not understand**. Security measures such as firewalls and VPNs are configuration layers on top of switching and routing. If you do not know how a frame or a packet travels through the network, you cannot decide where to filter it, or which part of it must be encrypted. That is why the first half of the lab is dedicated to infrastructure (switches, routers, VLANs, spanning tree and routing protocols), and the second half builds the security layers on top of it.

After completing all sessions, you should be able to:

- build, configure and troubleshoot a small hierarchical network with switches and routers, using static addressing, static routes and dynamic routing protocols;
- explain how Ethernet, ARP, IP and routing interact, and verify this by capturing and analyzing packets in Wireshark;
- design redundant Layer 2 networks and explain how Spanning Tree Protocol prevents loops and reacts to failures;
- segment a network with VLANs and 802.1Q trunks;
- configure zone-based firewall policies, and explain the difference between traffic to the router (input) and traffic through the router (forward), and between a denylist and an allowlist model;
- deploy and analyze a client-to-site VPN (WireGuard) and a site-to-site VPN (IPsec with IKEv2), and distinguish the inner from the outer packet;
- design a network with a DMZ, and validate it with Nmap and basic penetration-testing techniques;

2 Overview of lab sessions

This lab consists of 8 sessions of 4 hours, which can be divided into 2 parts, as shown in Figure 1.

The lab starts from a single device and gradually grows into a complete, secured company network. Every session builds on the previous one: the networks, tools, and concepts introduced early on return in almost every later lab.

- **Part 1 – Network infrastructure (sessions 1–4).** You learn how the devices work, how frames are switched, how packets are routed, how redundancy is provided with spanning tree, and

how a network is divided into VLANs. Packet captures play a central role: you do not just configure; you also verify what actually goes over the wire.

- **Part 2 – Network security (sessions 5–8).** You first control which traffic is allowed with firewalls, then protect traffic that crosses an untrusted network with WireGuard and IPsec, and finally combine everything in a company network with a DMZ that you defend and attack.

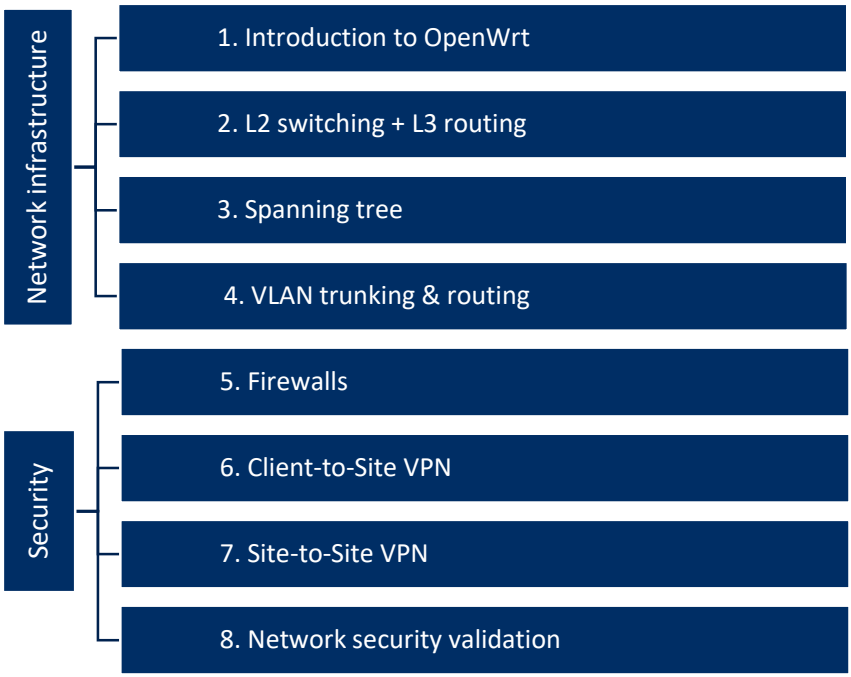


Figure 1: Overview of the eight lab sessions

3 Practicalities

All sessions are **physical, hands-on labs.** Together with your group, you will cable real switches, routers, and hosts, and every group works on its own physical network. This is a deliberate choice. Problems such as a wrong cable, a port in the wrong VLAN, or a forgotten firewall rule are exactly the problems you will encounter in practice, and they teach you to troubleshoot systematically.

In this lab, we will work with managed switches, such as the **NETGEAR GS108T-300PES** [1], and routers, like the **ASUS RT-AX53U** [2], both running **OpenWrt**. OpenWrt provides a flexible, Linux-based platform [3] that allows you to interact directly with routing, firewall, VLAN, and VPN configurations, giving insight into how real-world networks operate beyond vendor-specific interfaces. Each time a device is reset, it returns to a clean OpenWrt installation, so you can always start from a known state. By default, a device is reachable at 192.168.1.1, with user root and no password. This is acceptable in the isolated lab environment, but never in a real deployment.

Each lab guide contains exercises with questions. Think about a question and, where asked, make a prediction **before** you configure or measure; then check whether the result matches. Some questions require screenshots, captures, or command output.

The lab assistants are there to help you, but you are expected to first try to solve the problem yourself using all resources available to you, such as documentation, the internet, AI tools, and the course materials. For questions about the organization of the lab, contact the lecturers: Wout Van Uytsel (wout.vanuytsel@uantwerpen.be) and Femke Wouters (femke.wouters@uantwerpen.be).

4 Lab preparation

In order to maximize practical learning during the lab sessions, we kindly ask you to prepare for each lab session at home. The required preparation varies from session to session and is clearly described at the beginning of each lab guide.

At the start of every lab session (except Lab 1), you will find an overview of the topics you are expected to study, along with relevant references and resources. **Your preparation and understanding of these topics will be assessed at the beginning of each lab session**, and these assessments contribute to your final course grade.

5 Evaluation

Fout! Verwijzingsbron niet gevonden. summarizes the evaluation of the course, with division of theory and lab course, and subdivision. The laboratory (practicum) grade is composed of three elements:

- **Preparation tests (15%)** – Weekly individual multiple-choice tests via Blackboard assessing your readiness for the lab, including the required theoretical and practical knowledge.
- **Group grade (20%)** – A group score awarded during lab sessions when key checkpoints are validated by the lab assistant. Assessment covers correct implementation, conceptual understanding, technical execution, functionality, completeness, and achievement of the lab objectives.
- **Individual grade (15%)** – An individual score assigned by the instructor based on participation, contribution, communication, and professional attitude.

During the semester, you will complete **peer assessments** evaluating your own and your group members' contributions. These assessments will be used to adjust your individual lab grade and group grade based on effort, collaboration, technical contribution, reliability, and communication, ensuring that the final grade fairly reflects each student's contribution. This works with a multiplier concept between 0.8 and 1.2.

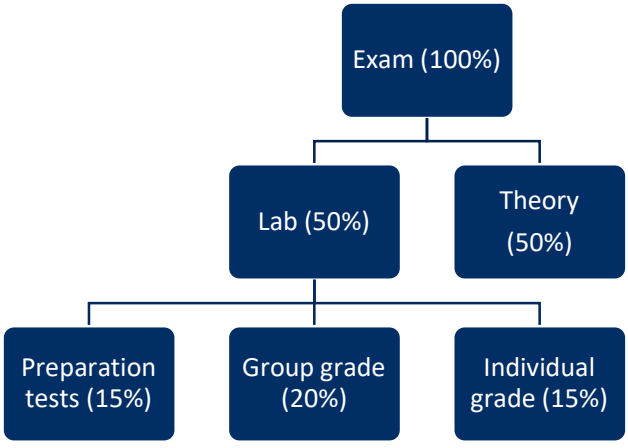


Figure 2: Course evaluation

It is important to note the following:

- You have to score at least 50% for both the lab and the theory.
- You have to attend 80% of the labs, which means you can only miss one opportunity.
- In case you do not pass a preparation test, you need to write a report about that lab session.